

Kantipur City College  
Department of Computer Engineering  
Semester IV/ Probability and Statistics  
Lab Sheet No.9

Title: Poisson Probability Distribution using EXCEL.

Objectives:

- i) To calculate Poisson probabilities.
- ii) To fit Poisson distribution

Theory:

**Poisson distribution**

A discrete random variable is said to have Poisson distribution with mean  $\lambda$  if

$$\bullet P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} \dots (1) \quad \text{for } x = 0, 1, 2, \dots$$

where,  $\lambda$  = mean or average number of occurrences called parameter.

$e = 2.7183$  is the base of natural logarithm

$x$  = number of occurrences of given event.

Symbolically we write  $X \sim P(\lambda)$ .

**Fitting of Poisson distribution:**

If a series of trials is repeated  $N$  times and satisfied the condition of Poisson distribution, then expected or theoretical frequency of  $x$  success is given by

$$f(x) = N p(x)$$

$$= N \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots$$

**Lab work 1:**

A random variable follows Poisson distribution with parameter  $\lambda = 1.5$ . Find the probabilities of  $X=2$ ,  $X < 5$  and  $x > 3$ ,  $X \leq 3$ , and  $3 < X < 8$ .

**Using Excel:**

|   | A            | B        | C | D                                | E |
|---|--------------|----------|---|----------------------------------|---|
| 1 | Lab work 1:  |          |   |                                  |   |
| 2 |              |          |   |                                  |   |
| 3 | $\lambda$    | 1.5      |   |                                  |   |
| 4 | $P(x=2)$     | 0.251021 |   | =POISSON(2,B1,0)                 |   |
| 5 | $P(x>3)$     | 0.065642 |   | =1-POISSON(3,B1,1)               |   |
| 6 | $P(x\leq 3)$ | 0.934358 |   | =POISSON(3,B1,1)                 |   |
| 7 | $P(x<5)$     | 0.981424 |   | =POISSON(4,B1,1)                 |   |
| 8 | $P(3<x<8)$   | 0.065473 |   | =POISSON(7,B1,1)-POISSON(3,B1,1) |   |
| 9 |              |          |   |                                  |   |

### Lab work 2:

Fit Poisson distribution for the following distribution:

|   |   |   |    |    |    |
|---|---|---|----|----|----|
| X | 4 | 3 | 2  | 1  | 0  |
| f | 2 | 3 | 19 | 65 | 11 |

### Using Excel:

|    |             |     |                    |                             |                            |               |
|----|-------------|-----|--------------------|-----------------------------|----------------------------|---------------|
| 12 | Lab work 2: |     |                    |                             |                            |               |
| 13 |             |     |                    |                             |                            |               |
|    |             |     | Expected frequency |                             | Rounded expected frequency |               |
| 14 | X           | f   | $f(x) = N p(x)$    | Formula                     |                            | Formula       |
| 15 | 4           | 2   | 2.602317991        | =\$B\$20*POISSON(A15,B\$23) | 3                          | =ROUND(C15,0) |
| 16 | 3           | 3   | 8.674393303        |                             | 9                          |               |
| 17 | 2           | 19  | 21.68598326        |                             | 22                         |               |
| 18 | 1           | 65  | 36.14330543        |                             | 36                         |               |
| 19 | 0           | 11  | 30.11942119        |                             | 30                         |               |
| 20 | Total       | 100 |                    |                             | 100                        |               |

### Assignments:

Fit a Poisson distribution and find i)  $P(X=3)$  ii)  $P(X>3)$  iii)  $P(X>=5)$  iv)  $P(4\leq x\leq 8)$

|   |      |      |      |     |    |    |   |   |   |
|---|------|------|------|-----|----|----|---|---|---|
| X | 0    | 1    | 2    | 3   | 4  | 5  | 6 | 7 | 8 |
| f | 2110 | 2610 | 1800 | 101 | 50 | 30 | 5 | 2 | 1 |